

MATLAB Scripts – Sets 8 and 12

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Set 8:

Given the surface charge density, $\rho_s = 2.0 \frac{\mu C}{m^2}$, existing in the region $r = 1.0 \text{ m}$, $0 < \phi < 2\pi$, $0 < \theta < \pi$ and is zero elsewhere (See Figure 8.2). Find analytically the energy stored in the region bounded by $2.0 \text{ m} < r < 3.0 \text{ m}$, $0 < \phi < 2\pi$, and $0 < \theta < \pi$. Write a MATLAB program to verify your answer.

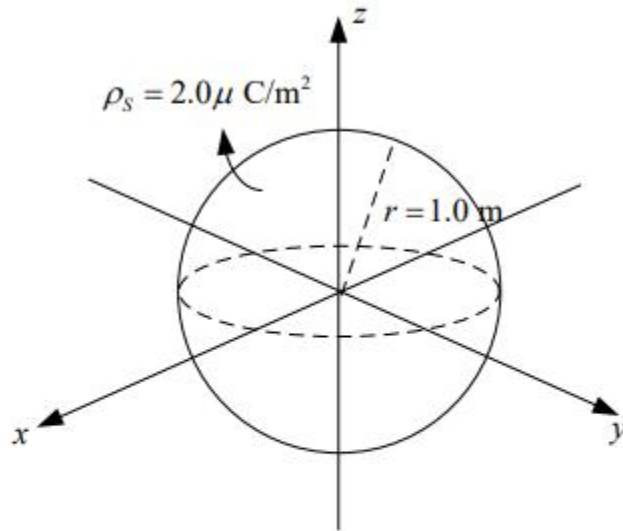


Figure 8.2 The surface charge density $\rho_s = 2.0 \mu \text{ C/m}^2$ at $r = 1.0 \text{ m}$.

Analytical Result:

$$\text{Given: } \epsilon_0 = 8.854 \times 10^{-12} \frac{\text{C}^2}{\text{Nm}^2}, \rho_s = 2.0 \frac{\mu \text{C}}{\text{m}^2}, r = 1.0 \text{ m}$$

Find the total charge:

$$Q = \rho_s(4\pi r^2)$$

$$Q = \rho_s[4\pi(1)^2]$$

$$Q = 4\pi\rho_s$$

Sub in ρ_s :

$$Q = 4\pi(2.0 \frac{\mu \text{C}}{\text{m}^2})$$

$$Q = 8\pi \times 10^{-6} \text{ C}$$

Find the electric field ($\vec{E}$):

For $r > 1.0 \text{ m}$, by Gauss's law the field is identical to that of a point charge Q at the origin:

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r$$

Magnitude:

$$E = |\vec{E}| = \frac{Q}{4\pi\epsilon_0 r^2} = \frac{8\pi \times 10^{-6}}{4\pi\epsilon_0 r^2} = \frac{2 \times 10^{-6}}{\epsilon_0 r^2}$$

Set up the energy integral:

$$W_E = \frac{1}{2} \iiint \epsilon_0 E^2 dv$$

In spherical coordinates, $dv = r^2 \sin \theta d\phi d\theta dr$

Sub in dv and E and integrate across $2.0 \text{ m} < r < 3.0 \text{ m}$, $0 < \phi < 2\pi$, and $0 < \theta < \pi$:

$$W_E = \frac{1}{2} \int_{r=2}^{r=3} \int_{\theta=0}^{\theta=\pi} \int_{\phi=0}^{\phi=2\pi} \epsilon_0 \left(\frac{2 \times 10^{-6}}{\epsilon_0 r^2} \right)^2 r^2 \sin \theta d\phi d\theta dr$$

$$W_E = \frac{1}{2} \int_2^3 \int_0^\pi \int_0^{2\pi} \epsilon_0 \left(\frac{4 \times 10^{-12}}{\epsilon_0^2 r^4} \right) r^2 \sin \theta d\phi d\theta dr$$

$$W_E = \frac{1}{2} \int_2^3 \int_0^\pi \int_0^{2\pi} \frac{4 \times 10^{-12}}{\epsilon_0 r^2} \sin \theta d\phi d\theta dr$$

$$W_E = \frac{4 \times 10^{-12}}{2\epsilon_0} \int_2^3 \int_0^\pi \int_0^{2\pi} \frac{1}{r^2} \sin \theta d\phi d\theta dr$$

$$W_E = \frac{2 \times 10^{-12}}{\epsilon_0} \int_2^3 \int_0^\pi \int_0^{2\pi} \frac{1}{r^2} \sin \theta d\phi d\theta dr$$

Solve energy integral:

$$W_E = \frac{2 \times 10^{-12}}{\epsilon_0} \int_2^3 \int_0^\pi \int_0^{2\pi} \frac{1}{r^2} \sin \theta d\phi d\theta dr$$

$$W_E = \frac{2 \times 10^{-12}}{\epsilon_0} \int_2^3 \int_0^\pi \left[\frac{1}{r^2} \sin \theta (\phi) \right]_{\phi=0}^{\phi=2\pi} d\theta dr$$

$$W_E = \frac{2 \times 10^{-12}}{\epsilon_0} \int_2^3 \int_0^\pi \left[\frac{1}{r^2} \sin \theta (2\pi) - \frac{1}{r^2} \sin \theta (0) \right] d\theta dr$$

$$W_E = \frac{2 \times 10^{-12}}{\epsilon_0} \int_2^3 \int_0^\pi \frac{2\pi}{r^2} \sin \theta d\theta dr$$

$$W_E = \frac{4\pi \times 10^{-12}}{\varepsilon_0} \int_2^3 \int_0^\pi \frac{1}{r^2} \sin \theta \, d\theta dr$$

$$W_E = \frac{4\pi \times 10^{-12}}{\varepsilon_0} \int_2^3 \left[\frac{1}{r^2} (-\cos \theta) \right]_{\theta=0}^{\theta=\pi} dr$$

$$W_E = \frac{4\pi \times 10^{-12}}{\varepsilon_0} \int_2^3 \left[\frac{1}{r^2} (-\cos \pi) + \frac{1}{r^2} (\cos 0) \right] dr$$

Sub in $\cos 0 = 1$ and $\cos \pi = -1$:

$$W_E = \frac{4\pi \times 10^{-12}}{\varepsilon_0} \int_2^3 \left[\frac{1}{r^2} (-(-1)) + \frac{1}{r^2} (1) \right] dr$$

$$W_E = \frac{4\pi \times 10^{-12}}{\varepsilon_0} \int_2^3 \left[\frac{1}{r^2} (1) + \frac{1}{r^2} (1) \right] dr$$

$$W_E = \frac{4\pi \times 10^{-12}}{\varepsilon_0} \int_2^3 \left(\frac{1}{r^2} + \frac{1}{r^2} \right) dr$$

$$W_E = \frac{4\pi \times 10^{-12}}{\varepsilon_0} \int_2^3 \frac{2}{r^2} dr$$

$$W_E = \frac{8\pi \times 10^{-12}}{\varepsilon_0} \int_2^3 \frac{1}{r^2} dr$$

$$W_E = \frac{8\pi \times 10^{-12}}{\varepsilon_0} \left(-\frac{1}{r} \right)_{r=2}^{r=3}$$

$$W_E = \frac{8\pi \times 10^{-12}}{\varepsilon_0} \left[-\frac{1}{3} - \left(-\frac{1}{2} \right) \right]$$

$$W_E = \frac{8\pi \times 10^{-12}}{\varepsilon_0} \left(-\frac{1}{3} + \frac{1}{2} \right)$$

$$W_E = \frac{8\pi \times 10^{-12}}{\varepsilon_0} \left(\frac{1}{6} \right)$$

$$W_E = \frac{8\pi \times 10^{-12}}{6\varepsilon_0}$$

$$\text{Sub in } \varepsilon_0 = \frac{10^{-9}}{36\pi}:$$

$$W_E = \frac{8\pi \times 10^{-12}}{6 \left(\frac{10^{-9}}{36\pi} \right)}$$

$$W_E = \frac{8\pi \times 10^{-12}(36\pi)}{6 \times 10^{-9}}$$

$$W_E = \frac{288\pi^2 \times 10^{-12}}{6 \times 10^{-9}}$$

$$W_E = 48\pi^2 \times 10^{-3}$$

$$W_E \approx 0.47374 \text{ J}$$

$\therefore$ the energy stored in a region bounded by $2.0 \text{ m} < r < 3.0 \text{ m}$, $0 < \phi < 2\pi$, and $0 < \theta < \pi$ with surface charge density, $\rho_s = 2.0 \frac{\mu\text{C}}{\text{m}^2}$, at $r = 1.0 \text{ m}$ is approximately 0.47374 J which was close to the result the MATLAB code generated verifying our results.

The MATLAB code generated an answer of $\approx 0.47376 \text{ J}$ which only has a 0.0042% percent error to the analytical result. To generate an even closer result, the number of discretization steps for the integration in the MATLAB code would be increased (equivalently, the step size would be decreased), which would improve accuracy at the cost of greater computational expense.

MATLAB Output to Verify Answers:

Command Window

```
The energy stored in the region is 0.47376 J
fx >>
```

Set 12:

A very long coaxial capacitor has an inner radius of $\rho_{inner} = 1.0 \times 10^{-3} \text{ m}$ and an outer radius of $\rho_{outer} = 5.0 \times 10^{-3} \text{ m}$. It is filled with a nonuniform dielectric characterized by $\epsilon_r = 10^3 \rho$. Find the capacitance of a 0.01 m long capacitor of this kind. Write a MATLAB program that finds the energy stored in this capacitor if the charge on the inner plate is $Q = 5.0 \times 10^{-9} \text{ C}$. Use the formula $W_E = \frac{Q^2}{2C}$ to evaluate the capacitance again and compare your results.

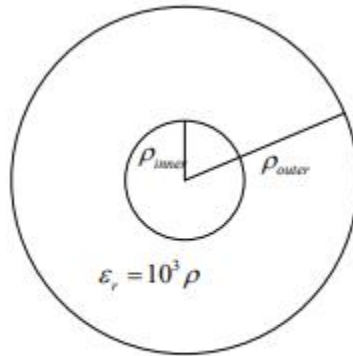


Figure 12.2 The cross section of the coaxial capacitor of the exercise of Set 12.

$$\text{Given: } \epsilon_0 = 8.854 \times 10^{-12} \frac{\text{C}^2}{\text{Nm}^2} = \frac{10^{-9} \text{ F}}{36\pi \text{ m}}, \rho_{inner} = 1.0 \times 10^{-3} \text{ m}, \rho_{outer} = 5.0 \times 10^{-3} \text{ m},$$

$$\epsilon_r = 10^3 \rho, Q = 5.0 \times 10^{-9} \text{ C}, h = 0.01 \text{ m}$$

Gauss's law for a coaxial structure:

$$\vec{D} = \frac{Q}{2\pi\rho h} a_\rho$$

$$\vec{E} = \frac{\vec{D}}{\epsilon_r \epsilon_0} = \frac{Q}{2\pi\epsilon_0(10^3\rho)\rho h} a_\rho = \frac{Q}{2\pi\epsilon_0 \cdot 10^3 h \cdot \rho^2} a_\rho$$

Finding Voltage:

Voltage between plates:

$$V = - \int_{\rho_{outer}}^{\rho_{inner}} \vec{E} d\rho = \int_{\rho_{inner}}^{\rho_{outer}} \frac{Q}{2\pi\epsilon_0 \cdot 10^3 h \cdot \rho^2} d\rho$$

$$V = \frac{Q}{2\pi\epsilon_0 \cdot 10^3 h} \int_{\rho_{inner}}^{\rho_{outer}} \frac{1}{\rho^2} d\rho$$

$$V = \frac{Q}{2\pi\epsilon_0 \cdot 10^3 h} \left(-\frac{1}{\rho} \right)_{\rho_{inner}}^{\rho_{outer}}$$

$$V = \frac{Q}{2\pi\epsilon_0 \cdot 10^3 h} \left(-\frac{1}{\rho_{outer}} - \left(-\frac{1}{\rho_{inner}} \right) \right)$$

$$V = \frac{Q}{2\pi\epsilon_0 \cdot 10^3 h} \left(-\frac{1}{\rho_{outer}} + \frac{1}{\rho_{inner}} \right)$$

Sub in values except Q:

$$V = \frac{Q}{2\pi \left(\frac{10^{-9}}{36\pi} \right) \cdot 10^3 (0.01 \text{ m})} \left(-\frac{1}{5.0 \times 10^{-3} \text{ m}} + \frac{1}{1.0 \times 10^{-3} \text{ m}} \right)$$

$$V = \frac{Q}{2\pi \left(\frac{10^{-9}}{36\pi} \right) \cdot 10^3 (0.01 \text{ m})} (800 \text{ m})$$

$$V = \frac{800Q}{2\pi \left(\frac{10^{-9}}{36\pi} \right) \cdot 10^3 (0.01 \text{ m})}$$

$$V = \frac{800Q}{\left(\frac{10^{-8}}{18} \right)}$$

Finding Capacitance:

$$C = \frac{Q}{V} = \frac{Q}{\frac{800Q}{\left(\frac{10^{-8}}{18} \right)}} = \frac{\left(\frac{10^{-8}}{18} \right)}{800}$$

$$C \approx 6.9444 \times 10^{-13} \text{ F} \approx 0.694 \text{ pF}$$

Energy Stored:

$$W_E = \frac{Q^2}{2C}$$

Sub in values:

$$W_E = \frac{(5.0 \times 10^{-9} \text{ C})^2}{2(6.9444 \times 10^{-13} \text{ F})}$$

$$W_E \approx 1.80 \times 10^{-5} \text{ J} \approx 18.0 \text{ } \mu\text{J}$$

∴ the capacitance of a 0.01 m long capacitor of this kind is approximately 0.694 pF or $6.9444 \times 10^{-13} \text{ F}$ and the energy stored in this capacitor if the charge on the inner plate is $5.0 \times 10^{-9} \text{ C}$ is approximately 18.0 μJ or $1.80 \times 10^{-5} \text{ J}$. This was close to the result the MATLAB code generated verifying our results. The MATLAB code generated an answer of $\approx 6.9456 \times 10^{-13} \text{ F}$ for capacitance which only has a 0.017% percent error to the analytical result. For the energy stored in the capacitor, the MATLAB code generated an answer of $\approx 1.7997 \times 10^{-5} \text{ J}$ which only has a 0.017% percent error to the analytical result. To generate an even closer result, the number of discretization steps for the integration in the MATLAB code would be increased (equivalently, the step size would be decreased), which would improve accuracy at the cost of greater computational expense.

MATLAB Output to Verify Answers:

```
Command Window
The capacitance of a capacitor of this kind with 0.01 m length is 6.945592e-13 F
The energy stored in this capacitor is 1.799703e-05 J
fx >> |
```